



Team members' names:

1. Aakash Nihalaney
2. Anrunya Patole
3. Avi Gupta
4. Harsh Sahu
5. Kshama Purohit
6. Om Kapdoskar

1. Big Idea (write a sentence or two below describing the overall reason for your project):

The goal of this project is to conduct a “Stress Test” of Virginia Tech’s infrastructure to evaluate how housing, dining, transportation, and student services are keeping up with a rapidly growing student population. By visualizing wait times, reliability, and resource availability, we aim to identify structural pressures on the “Hokie Experience”.



2. Your data collection table (fill out this table, use the template on Canvas to help you). Think of each row in the table as a dashboard in your Tableau data story. You can add rows to the table.

Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
Are dining services able to handle peak student demand on campus?	Wait times at dining halls, frequency, of switching to off-campus food or Grab & Go options, peak hours.	Average wait time, frequency ratings, average wait time comparison between GrubHub users and non-users, off-campus dining willingness score by class year .	Survey responses, Dining staff interviews..	Online survey responses, In-person interview with dining hall staff, informal conversions with students in waiting line.	Project team members.	Data collected from VT students through online responses, VT Dining halls.
Which student demographics experience the most infrastructure stress?	Student status (undergraduate, graduate, professor), gender, residence location, time spent on campus, college/major.	Distribution of stress levels by student category, comparison of responses across demographic groups, average infrastructure stress score	Survey responses; Student interviews; Peer conversions.	Online survey responses; In-person interviews with students of different majors, years, and living situations.	Project team members.	Data collected from VT students through online responses; Collected on campus (libraries, dorms, student lounges)

¹ See the handout on Canvas for an example of what should go in each field.

² Data collectors = where you can get the data from (e.g., which websites) and who is going to be responsible for collecting that portion of the data.



Pamplin College of Business
Business Visualization Analytics/ Spring 2026

Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
		across all respondents.				throughout the semester
How do infrastructure constraints impact academic outcomes?	Number of missed classes due to commuting issues, class schedule constraints, stress levels related to campus logistics.	Frequency of missed classes; correlation between commute time and reported stress or academic disruptions.	Survey responses; Student interviews, professor/TA observations	Online survey responses; Short in-person interviews with students and TAs about attendance and commuting challenges	Project team members.	Data collected from VT students through online responses; Collected near classrooms and transit stops before/after class times
Are Virginia Tech's physical and digital student services accessible enough to support a growing student population?	Frequency of successful appointment bookings at service buildings, which library resources are hardest to access, how often students visit the gym and at what times, overall	Booking success rates across service buildings, distribution of gym crowding by time slot, average infrastructure stress score across demographics	Survey responses, VT Student Services websites, gym attendance observations	Online survey responses, informal conversations with students near service buildings and the gym	Project team members	Data collected from VT students through online responses, collected on campus near service buildings, the library, and gym facilities throughout the semester



Pamplin College of Business
Business Visualization Analytics/ Spring 2026

Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
	infrastructure stress rating by student group					
How effectively does Virginia Tech's transit infrastructure support students getting to class on time?	Where students commute from, mode of transport used, time taken to get from campus to class, classes missed due to transit or parking issues, BT Transit satisfaction by commute area	Average commute to class time by arrival window, frequency of missed classes due to transit issues, satisfaction score with BT Transit broken down by commute origin	Survey responses, Blacksburg Transit route information at ridebt.org , VT parking and transportation at parking.vt.edu	Online survey responses, short in-person interviews with students at bus stops and parking areas	Project team members	Data collected from VT students through online responses, collected near BT bus stops and campus parking areas before and after peak class times
How well are Virginia Tech's digital platforms serving the needs of a growing and increasingly dependent student body?	Satisfaction ratings for each digital platform, which platform students most want redesigned, average rating across all services, open ended feedback	Satisfaction score per platform on a scale of 1 to 5, count of students selecting each platform for redesign, frequency of	Survey responses, VT IT Strategic Plan at it.vt.edu , student open ended written feedback	Online survey responses, in-person conversations with students about their experiences using platforms like Hokie SPA and Canvas	Project team members	Data collected from VT students through online responses, collected in libraries, computer labs, and student lounges where



Inquiry Question (Curiosity) ¹	Information Needed to Answer Inquiry Question	Measurement Indicator	Data Sources	Data Collection Methods	Data Collectors ²	Timing and Location of Data Collection
	on what students would change	negative sentiment words in open ended responses				students actively use these platforms
Which dining halls on campus are absorbing the most student demand and are they distributed in a way that matches where students actually spend their time?	Student dining hall preferences by location, which halls are most popular at different times of day, breakdown of dining preferences by student status, average wait time for GrubHub users vs non-users, off-campus willingness score by class year	Count of students preferring each dining location, most popular hall by time slot, distribution of preferences across student status groups, average wait time for GrubHub users vs non-users, off-campus willingness score by class year	Survey responses, VT Dining Services at dining.vt.edu, campus map data	Online survey responses, informal conversations with students at dining locations during peak and off peak hours	Project team members	Data collected from VT students through online responses, collected at and near dining halls across campus during breakfast, lunch, and dinner windows throughout the semester

3. Describe how you will analyze (make meaning from) the collected data (write your plan below):

Think about what relationships you hope to uncover in the data and what different audiences you expect to present your data to.



- The results of the survey will be analyzed for patterns among student experiences across different infrastructure systems through descriptive statistics and visual analytics in Tableau. First, summary statistics will be calculated for key variables, including average wait times for dining, average time spent commuting, student satisfaction with digital services, and average instances of missing class. These statistics will then be visualized in dashboards that show bottlenecks in university operations, including heat maps for wait times, maps for commuting areas, and stacked bar charts for student experiences across different demographics. Additionally, correlation analysis will be conducted on variables, including student stress levels and instances of missing class, given longer commutes. The results will be presented primarily for university stakeholders, including students and faculty members who have an interest in university planning, as they will be able to identify areas of improvement for university infrastructure as it continues to expand.